Rutvik Belsare

Bangalore, India belsarerutvik@gmail.com +91 72182 10955

Research Portfolio LinkedIn

Prospective Research Focus: Urban Informatics, Transit Accessibility, & Data-Driven Spatial Decision Support. Seeking an MPhil / Masters by Research degree. Combining a rigorous Bachelor of Planning background (COEP) with enterprise computational experience (o9 Solutions) to build scalable quantitative models for public transport integration and resilient urban spatial networks.

EDUCATION

2020 – 2024
Pune, India

COEP Technological University (formerly College of Engineering Pune)

Bachelor of Technology in Planning (B.Plan / B.Tech Planning)

- **Academic Standing:** CGPA 8.42/10.0 (~80% aggregate score; Equivalent to Australian First Class Honours / High Distinction standard). Degree awarded through Savitribai Phule Pune University.
- **Thesis / Capstone Distinction:** Awarded Grade AB in Planning Thesis I & II (Focus: First & Last Mile Multimodal Integration for Metro Rail).
- **Supplementary Coursework:** Minor Certification in Computer Technology (Data Structures, Python, Systems Analysis).

2020
Nagpur

Shivaji Science College, Nagpur

Senior Secondary Education (Class XII), Electronics · Percentage: 80.00%

2018
Nagpur

Saraswati Vidyalaya, Nagpur

Secondary Education (Class X) · Percentage: 93.20%

PEER-REVIEWED PUBLICATIONS & RESEARCH OUTPUTS

1. **Optimizing Urban Mobility: A Study of First and Last Mile Connectivity for Pune Metro** BOOK CHAPTER

   Rutvik Belsare and Dr. Arati S. Petkar.
   Published in Springer Book Series (Conference Proceedings in Urban & Transportation Infrastructure), 2026. Presented at International Conference, July 2024.

   - Formulated an empirical commuter transit model evaluating catchment accessibility, feeder modes, and transfer barriers surrounding Pune Metro stations.

2. **Urban Mobility Infrastructure: Bus Service, Smart Parking and Public Bike Sharing, Nashik Smart City** NATIONAL COMPENDIUM

   Chidambari Kulkarni, Aryan Mande, and Rutvik Belsare (Advisors: Dr. Arati S. Petkar, Ms. Lisha C. Bendre).
   Published in SAAR Compendium of Landmark Urban Case Studies (Ministry of Housing and Urban Affairs / NIUA), March 2023.

   - Evaluated multi-modal integration metrics across 75+ national smart city projects for systemic performance assessment.

3. **Area Appreciation of Narendra Nagar: A Spatial Case Study of Nagpur** CONFERENCE PROCEEDINGS
Rutvik Belsare and Pooja Gandhile.
Published in Peer-Reviewed Proceedings, BNCA National Conference on Built Environment and Beyond, Oct 2022.
- Executed spatial syntax and socioeconomic field surveys (200+ households) analyzing urban morphology and infrastructure deficits.

4. **A Designed Model of Neighborhood in Pune by Analyzing Concepts of Sustainability and Affordable Housing** CONFERENCE PROCEEDINGS
Rutvik Belsare, Aarya Watane, and Shubhangi Thakre.
Published in Peer-Reviewed Proceedings, BNCA National Conference on Built Environment and Beyond, Oct 2022.

5. **Achieving Balanced Development Through Industrial Growth: A Case of Ratnagiri Region** CONFERENCE PROCEEDINGS
Aarya Watane, Rutvik Belsare, Dr. Arati S. Petkar, and Dr. S. G. Sonar.
National Conference on Regional Development Issues, Gokhale Institute of Politics & Economics / Regional Science Association, Dec 2023.

RESEARCH EXPERIENCE

Jun 2022 – Dec 2022

University of California, Irvine (UCI) – Dept. of Urban Planning & Public Policy

Graduate Research Assistant (Collaborated with PhD Candidate)

- Executed field outreach, primary survey design, and qualitative interview coding evaluating post-disaster community well-being following the 2021 Western Maharashtra floods.
- Engineered bilingual survey translation frameworks and conducted secondary demographic data synthesis to evaluate regional climate resilience and institutional response metrics.

INDUSTRY & APPLIED ANALYTICS EXPERIENCE

Aug 2024 – Present
Bangalore, India

o9 Solutions, Inc.

Senior Functional Consultant / Application Engineer II (Enterprise Planning Systems)

- Engineered algorithmic decision-support and scenario-planning modules for a global Fortune 500 enterprise using advanced relational data models.
- Authored custom Python plugins for real-time Key Performance Indicator (KPI) calculation, automated anomaly detection, and root-cause diagnostic workflows.
- Optimized large-scale database queries and user interface workflows, leading technical audits for system performance and scalable deployment.

May 2023 – Jul 2023
Pune, India

Bajaj Allianz General Insurance

ESG & Sustainability Planning Research Intern

- Synthesized global Environmental, Social, and Governance (ESG) compliance frameworks, authoring a benchmark report on sustainable corporate infrastructure standards.

ACADEMIC PLANNING STUDIOS & APPLIED PROJECTS

2020 – 2024

Comprehensive Studio Portfolio

COEP Technological University

- **First & Last Mile Transit Access Study (B.Plan Thesis):** Spatial accessibility modelling, modal split analysis, and infrastructure redesign recommendations for Pune Metro rail corridors.
- **Ratnagiri Regional Development Plan (Regional Studio):** Modelled economic drivers, land-use allocation, and industrial corridors for regional spatial balance. Role: Studio Lead (20-student cohort).
- **Kasba Peth Local Area Plan (Urban Design Studio):** Inner-city urban renewal, heritage corridor conservation, and pedestrianisation schemes for high-density historic precincts.
- **Vadgaon Budruk Junction & Traffic Redesign (Transport Studio):** Intersection geometric redesign, signal timing optimisation, and pedestrian safety infrastructure proposal for Dhobi Ghat junction.

RESEARCH & TECHNICAL METHODOLOGIES

SPATIAL & GIS	ArcGIS, QGIS, Spatial Syntax, Land-Use/Transport Modeling, Satellite Remote Sensing
QUANTITATIVE & CODE	Python (Pandas, NumPy, GeoPandas), SQL, R (Basic), Mathematical Optimization
DESIGN & CAD	AutoCAD, SketchUp, Adobe Creative Suite (Photoshop, Illustrator), Figma
RESEARCH METHODS	Household Survey Design, Sampling & Stratification, Qualitative Coding, Statistical Hypothesis Testing

HONOURS, AWARDS & INSTITUTIONAL LEADERSHIP

- **1st Runner-Up, TCS Sustainathon (2021):** National competition on innovative technological solutions for urban sustainability and resource management.
- **First Position, Utopia (MindSpark 2023):** National-level urban planning design competition evaluating resilient city concepts.
- **Semi-Finalist, Prajatantra Policy Presentation (2022):** Presented policy recommendations on Municipal Finance Reform in urban local bodies.
- **Overall Head of Finance & Marketing, Pune StartUp Fest (2023):** Managed budgeting, corporate outreach, and financial operations for central university initiative.
- **Student Training & Placement Coordinator & Class Representative:** Elected representative across academic years 2020–2022 for the Department of Planning.
- **NPTEL Certificate – Introduction to Research (Score: 68%):** Formal training in literature review, ethics, and research methodology.

ACADEMIC REFEREES

Dr. Arati S. Petkar
Associate Professor & B.Tech Thesis Supervisor
Department of Civil Engineering & Planning, COEP Technological University, Pune, India
Email: asp.civil@coeptech.ac.in

Official Transcripts & Additional Academic References Available on Request